

Entropy 2

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Exercise 1.

Let $\mathcal{F}$ be a collection of subsets of $[n]$. Denote p_i the fraction of subsets in $\mathcal{F}_i$ that contain i . Prove that

$$|\mathcal{F}| \leq \prod_i p_i^{-p_i} (1 - p_i)^{-(1-p_i)}$$

Solution 1.

We have X be a random subset of $[n]$ in $\mathcal{F}$ and X_i be the indicator of the event $i \in X$. We have

$$\begin{aligned} \log_2 |\mathcal{F}| = H(X) &= \sum_i H(X_i | X_1, \dots, X_{i-1}) && \text{by Chain rule} \\ &\leq \sum_i H(X_i) && \text{by dropping the conditioning} \\ &= \sum_i -p_i \log_2 p_i - (1 - p_i) \log_2 (1 - p_i) \end{aligned}$$

Hence, $|\mathcal{F}| \leq \prod_i p_i^{-p_i} (1 - p_i)^{-(1-p_i)}$ as desired.

Exercise 2.

Let X, Y be random integer-valued variables.

1. Assume that X and Y are independent. Show that $H(X + Y) \geq \max(H(X), H(Y))$
2. If X and Y are no longer independent, can we have $H(X + Y) < \min(H(X), H(Y))$?

Solution 2.

1. $H(X + Y) \geq H(X + Y | Y) = H(X | Y) = H(X)$ because X and Y are independent.
2. Let X be a non-constant random variable and $Y = 1 - X$. Then $X + Y$ is constant, so $H(X + Y) = 0$, but $H(Y) = H(X) > 0$.

Exercise 3.

Let $\mathcal{G}$ be a family of graphs on $2n$ labelled vertices, such that for each $G_i, G_j \in \mathcal{G}$, the graph $G_i \cap G_j$ contains a perfect matching. Show that $|\mathcal{G}| \leq 2^{\binom{2n}{2} - n}$ and that this bound is tight.

Solution 3.

We view the graphs of $\mathcal{G}$ as subsets of edges of K_{2n} . For every vertex u of K_n , let A_u be the set of edges incident to u . For every u , we have $|A_u| = 2n - 1$. The sets $\mathcal{G}_{|A_u} = \{G \cap A_u : G \in \mathcal{G}\}$ form an intersecting family: for each $G_i, G_j \in \mathcal{G}$, $(G_i \cap A_u) \cap (G_j \cap A_u)$ contains the edges of $G_i \cap G_j$ that are incident to u and there is at least one such edge because $G_i \cap G_j$ contains a perfect matching. Thus, $|\mathcal{G}_{|A_u}| \leq 2^{|A_u|-1} = 2^{2n-2}$. Moreover, each edge of K_{2n} belongs to exactly two sets A_u . So by the combinatorial version of Shearer's inequality,

$$\begin{aligned} |\mathcal{G}| &\leq \sqrt{\prod_u |\mathcal{G}_{|A_u}|} \\ &\leq (2^{2n-2})^{2n/2} = 2^{\binom{2n}{2} - n} \end{aligned}$$

This bound is tight and attained by the following family $\mathcal{G}$. Let M be a perfect matching of K_{2n} and let $\mathcal{G}$ be the set of all graphs that contain all edges of M . Clearly, the intersection of every pair of graphs of $\mathcal{G}$ contains a perfect matching as it contains M , and we have $|\mathcal{G}| = 2^{\binom{2n}{2} - n}$.

Exercise 4.

Let H be a 5-cycle with vertices labelled from 1 to 5. Let G be a graph on n vertices with vertices labelled by $[n]$ and m edges. Recall that an embedding of H in G is a map $\phi : [5] \rightarrow [n]$ such that for every $ij \in E(H)$, $\phi(i)\phi(j) \in E(G)$.

Prove that the number of embeddings of H in G is at most $(2m)^{5/2}$.

Solution 4.

Let X be a random uniform embedding of H in G . More precisely, X is a 5-tuple in which each X_i is the image of i in G . For every edge uv of H , $H(X_u, X_v) \leq \log 2m$ by uniform bound, since G has m edges.

By Shearer's lemma, $2H(X) = 2H(X_1, \dots, X_5) \leq \sum_{uv \in E(H)} H(X_u, X_v) \leq 5 \log 2m$. Hence the number of embeddings of G is $2^{H(X)} \leq (2m)^{5/2}$.

Exercise 5.

Let A, B, C be finite subsets of $\mathbb{R}$. Prove that

$$|A + B + C|^2 \leq |A + B| \cdot |A + C| \cdot |B + C|$$

where $A + B = \{a + b : a \in A, b \in B\}$.

Solution 5.

Exercise 6. Submodularity and Shearer for sums of independent random variable

Let $I(X; Y) = H(X) + H(Y) - H(X, Y)$ be the mutual information between X and Y .

Data processing Prove that for every deterministic function f , $I(f(X); Y) \leq I(X, Y)$

Independence Prove that if Z is independent from (X, Y) , $I((X, Z); Y) = I(X; Y)$

Submodularity Let X, Y, Z be mutually independent random integers. Prove that $H(X + Y) + H(Y + Z) \geq H(X + Y + Z) + H(Y)$

Shearer Let X, Y, Z be mutually independent random integers. Prove that $2H(X + Y + Z) \leq H(X + Y) + H(X + Z) + H(Y + Z)$.

Solution 6.

We first prove the data processing inequality:

$$\begin{aligned} I(X; Y) - I(f(X); Y) &= H(X) - H(f(X)) - (H(f(X), Y) - H(X, Y)) \\ &= H(Y) - H(Y|f(X)) \end{aligned}$$

Assume Now that Z is independent from (X, Y) . We have

$$\begin{aligned} I((X, Z); Y) - I(X; Y) &= H(X, Z) - H(X, Z|Y) - (H(X) - H(X|Y)) \\ &= H(X, Z) - H(X) - (H(X, Z|Y) - H(X|Y)) \\ &= H(Z|X) - H(Z|X, Y) = 0 \end{aligned}$$

For independent variables X and Y , we have $I(X + Y; Y) = H(X + Y) - H(X + Y|Y) = H(X + Y) - H(X)$. Thus,

$$\begin{aligned} H(X + Y) + H(Y + Z) - H(X + Y + Z) - H(Y) &= I(X + Y; Y) - I(X + Y + Z; Y + Z) \\ &\geq I(X + Y; Y) - I(X + Y, Z; Y) && \text{by data processing} \\ &= 0 && \text{by independence} \end{aligned}$$

Finally, the Shearer inequality follows easily from submodularity:

$$\begin{aligned} 2H(X + Y + Z) - (H(X + Y) + H(Y + Z) + H(X, Z)) &= H(Y + Z) - H(Y) - H(Z) \\ &\leq H(Y, Z) - H(Y) - H(Z) \\ &\leq 0 \end{aligned}$$

Exercise 7.

A Sudoku of size n is an $n^2 \times n^2$ array $(s_{ij})_{i,j \in [n]}$ such that each line, each column and each block $(s_{ij})_{pn < i \leq (p+1)n, qn < j \leq (q+1)n}$ for some $p, q \in [n]$ contains every number from $[n^2]$.

1. Using the Poisson paradigm, guess an upper bound on the number of Sudokus of size n .
2. Prove it using the entropy method.

Solution 7.

1. We sample independently uniformly at random a number $X_{i,j} \in [n^2]$ in each cell. For each line, column or block, the expected number of times a fixed number appears is 1. Hence, by the Poisson paradigm, for each cell of coordinate (i, j) , the probability that $X_{i,j}$ appears only once in its line, column AND block, is $e^{-3} + o(1)$. If all these events were independent, then the probability that X is a sudoku would be $(e^{-3} + o(1))^{n^4}$. Hence, the number of Sudoku's of size n should be around

$$\left(\frac{n^2}{e^3 + o(1)} \right)^{n^4}$$

.

2. This proof is highly similar to the proof for Latin squares from the previous exercise sheet.